Ken Tanaka Hernández

Computational & Mathematical Physicist | High-Performance Computing & AI

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"We are a way for the cosmos to know itself." — Carl Sagan (1980)

Professional Summary

Computational and mathematical physicist specialized in theoretical and numerical modeling of complex stochastic dynamical systems across scales. Research foundation uniting **statistical physics, high-dimensional statistical inference, and large-scale scientific computing**. Proven expertise in **HPC, Big Data, and AI** by architecting and deploying **scalable, distributed systems** for high-throughput research and enterprise workloads, spanning **multi-node GPU Deep Learning pipelines on Tier-0 supercomputers (CINECA Leonardo)**.

Experience

Istituto Nazionale di Oceanografia e di Geofisica Sperimentale (OGS) & Università degli Studi di Trieste (UniTS) — Trieste, Italy

PhD Researcher — Nov 2023 – Present

- Architected and deployed automated **ML** seismic data processing pipelines on Italy's **Tier-0 CINECA Leonardo supercomputer** with dynamic job submission across **32+ cores/node** using **SLURM, MPI, and multi-node GPU parallelization**, achieving **94% accuracy** on high-throughput production workloads.

International Centre for Theoretical Physics (ICTP) — Trieste, Italy

Research Assistant — Apr 2023 – Aug 2023

- Investigated rank-one estimation in the spiked Wigner model under Prof. Jean Barbier; developed a **prior-independent scheduler** for **Approximate Message Passing (AMP)** Onsager corrections to characterize Bayes-optimal signal recovery under structured noise.

ORACLE de México, SA de CV — Guadalajara, Mexico

Software Developer II — Jul 2020 – Aug 2022

- Architected and deployed a **production backend** and customer-facing API for automated analytics and diagnostic patch recommendation workflows.
- Mentored junior developers and established engineering **code review practices**.

Hewlett Packard (HP) Labs — Guadalajara, Mexico

Intern, Modeling & Numerical Simulation — Jun 2019 – Jun 2020

- Developed thermal-process simulations for Multi Jet Fusion 3D printing technology.
- Validated computational models against experimental materials-science benchmarks.

Universidad de Guadalajara (UdeG) — Guadalajara, Mexico

Research Assistant — Apr 2018 – Jun 2020

- Conducted **computational physics** and **Markov Chain Monte Carlo (MCMC) simulations** for phase transition analysis and emergent phenomena in complex systems.
- Researched the Totally Asymmetric Simple Exclusion Process (**TASEP**) and Global Magnetization Conserved Spin Exchange Dynamics under Dr. Soham Biswas.

Universidad Autónoma de Nuevo León (UANL) — Monterrey, Mexico

Research Assistant — May 2018 – Jul 2018

- Researched computational algorithms for the local detection of community structures in complex networks under Dr. Satu Elisa Schaeffer.

Education

Università degli Studi di Trieste (UniTS) Trieste, Italy

PhD in Applied Data Science & Artificial Intelligence (ADSAI) Nov 2023 – Present
Built **distributed training/inference pipelines** across **multi-node GPU clusters (MPI/CUDA)** for large seismic datasets; engineered a custom validation framework using **bipartite optimization** and weighted similarity scoring.
Expected graduation: Oct 2026. Completed concurrently with MHPC studies.

Scuola Internazionale Superiore di Studi Avanzati (SISSA) Trieste, Italy

Master in High Performance Computing (MHPC) Sep 2023 – Aug 2024
Developed an **end-to-end ML pipeline** for real-time earthquake detection; architected **HPC-optimized workflows** with **Transformers and CNNs** to ingest continuous data streams at scale.
Graduated with honors. Completed concurrently with PhD studies.

International Centre for Theoretical Physics (ICTP) Trieste, Italy

Postgraduate Diploma in Quantitative Life Sciences (QLS) Sep 2022 – Aug 2023
Completed coursework in Universal Biology and Active Matter, Probability and Information Theory, Mathematical Economics and Game Theory, Computational Neuroscience, and Reinforcement Learning.

Universidad de Guadalajara (UdeG) Guadalajara, Mexico

BSc. Physics Aug 2016 – Dec 2021
Simulated complex social systems with **1D Ising spin models** integrating Kawasaki and Schelling dynamics; quantified critical exponents via **finite-size scaling** ($z \approx 2.48$, $\lambda \approx 1.15$, $\theta \approx 0.46$).
Graduated with postgraduate studies. Completed concurrently with BEng studies.

Instituto Tecnológico y de Estudios Superiores de Occidente (ITESO) Guadalajara, Mexico

BEng. Nanotechnology Aug 2016 – Dec 2018
Coursework concurrently with BSc studies.

Publications

Insights into a Deep Learning Workflow for Automatic Detection of Seismic Signals at Mount Etna Volcano

Carducci, A.; Cocina, O.; Sciotto, M.; Cannata, A.; Di Gioia, S.; Vuan, A.; Saraò, A.; **Tanaka Hernández, K.**; Sugan, M. *Seismological Research Letters*, 2026. DOI: 10.1785/0220250431.

Unpublished / Forthcoming

Evaluating an AI-enabled HPC workflow for seismic monitoring in the Southeastern Alps

Tanaka Hernández, K.; Münchmeyer, J.; Magrin, A.; Di Gioia, S.; Vuan, A.; Picozzi, M.; Rodriguez Garcia, A.; . . . ; Sugan, M. *Manuscript in preparation.*

Comparative Evaluation of Clustering Algorithms for Seismic Event Association: Assessing their suitability for reliable event association

Tanaka Hernández, K., Rodriguez Garcia, A., Vuan, A. *Manuscript in preparation.*

Datasets

Revised OGS Seismic Catalog for Northern Italy (2005–2024): Quality-Controlled, Homogeneous Dataset for Seismological and Machine Learning Applications

Magrin, A.; **Tanaka Hernández, K.**; Sugan, M. *Zenodo*, 2026. Dataset, version 1.0. DOI: 10.5281/zenodo.22106548.

Honors & Awards

2023–2024	**TeRABIT Scholarship** — **HPC** training and Earth-science research	OGS
2022–2023	**Postgraduate Diploma Fellowship** — Competitive **Theoretical Physics** scholarship	ICTP
2020	**3rd Place, National Poster Presentation Contest** — Research poster competition	HP

Conferences

Performance assessment of Deep Learning picking models at Mount Etna volcano

Carducci, A., Cocina, O., Sciotto, M., Cannata, A., Di Gioia, S., Vuan, A., Saraò, A., **Tanaka Hernández, K.**, and Sugan, M.: *EGU General Assembly 2026*, Vienna, Austria, 3–8 May 2026, EGU26-11585, DOI: 10.5194/egusphere-egu26-11585.

Benchmark tests of Deep Learning pickers at Mount Etna volcano

Carducci, A.; Cocina, O.; Sciotto, M.; Cannata, A.; Di Gioia, S.; Vuan, A.; Saraò, A.; **Tanaka Hernández, K.**; Sugan, M. (2026). *GNGTS – Gruppo Nazionale di Geofisica della Terra Solida, 44th National Conference of the GNGTS*, 10–13 February 2026, Udine, Italy.

Artificial Intelligence (AI) and Machine Learning (ML) on Big Data seismology

Tanaka Hernández, K.; Münchmeyer, J.; Magrin, A.; Di Gioia, S.; Rodriguez Garcia, A.; Sugan, M. (2026). *GNGTS – Gruppo Nazionale di Geofisica della Terra Solida, 44th National Conference of the GNGTS*, 10–13 February 2026, Udine, Italy.

Artificial Intelligence (AI) and Machine Learning (ML) on Big Data Seismology

Tanaka Hernández, K. (2025). *24th Mexican International Conference on Artificial Intelligence (MICAI 2025) Doctoral Consortium*, 3–7 November 2025, Guanajuato, Mexico.

Selected Projects

Automated Seismic Data Processing Pipeline

End-to-end HPC pipeline for seismic event detection: waveform acquisition, ML-based phase picking (SeisBench), octree-based association (PyOcto), and 3D hypocenter location (NonLinLoc). Dynamic SLURM job generation with template-based configuration and GNU Make orchestration.

Event Matching & Validation Framework

Novel validation system using **Hungarian algorithm-based bipartite optimization** with custom weighted similarity metrics (location, magnitude, origin time). Enables rigorous comparison against authoritative catalogs (INGV). Open-source contribution for reproducible seismology research.

High-Dimensional Statistical Inference & Approximate Message Passing

Investigated rank-one estimation in spiked Wigner matrix models under Prof. Jean Barbier at ICTP. Engineered a prior-independent Onsager reaction scheduler for Approximate Message Passing (AMP), establishing recovery thresholds and analyzing fundamental algorithmic phase transitions under structured noise.

Additive Manufacturing Thermal Simulator

High-fidelity **finite element simulation** of powder bed fusion processes. Validated against experimental thermal imaging data from HP's 3D Multi Jet Fusion production line.

Global Magnetization Conserved Spin Exchange Dynamics as a Social Dynamics Simulation Engine

Monte Carlo framework for studying segregation and opinion dynamics in complex networks. Implemented Ising-based models with Kawasaki-Schelling hybrid dynamics and characterized critical exponents via finite-size scaling. Research supervised by Dr. Soham Biswas.

Totally Asymmetric Simple Exclusion Process (TASEP)

Simulated established TASEP results and modeled the effects of a traffic light using **Fortran and GNUPlot**, under the supervision of Dr. Soham Biswas.

Local Detection of Community Structures

Introductory research during a Research Assistantship at Universidad Autónoma de Nuevo León (UANL), on local detection of community structures in complex networks under Dr. Satu Elisa Schaeffer.

Technical Skills

Theory & Modeling	**Statistical Physics & Nonequilibrium Dynamics**, High-Dimensional Inference, Complex Systems, Monte Carlo Methods, Approximate Message Passing (AMP)
HPC & Systems	**Tier-0 Supercomputing (Leonardo)**, **CUDA, MPI, OpenMP, SLURM**, Multi-Node GPU Scaling, Distributed Computing, Parallel Optimization
Programming & ML	**Python** (PyTorch, CuPy, Dask, NumPy, SciPy), **C/C++**, SQL (Oracle, PostgreSQL), Bash, **Transformers, CNNs, PINNs**
Infrastructure	Linux, Docker, Git, CI/CD, Pipeline Orchestration (**GNU Make, Hydra**)
Languages	Spanish (native), English (C1), Italian (A2), Japanese (conversational), Chinese (A1)

Volunteering

ISC High Performance Science and Technology

Student Volunteer Jun 2025
Engaged with leading minds in high performance computing, including renowned scientists, early-career researchers, and representatives of organizations advancing HPC.

Université Grenoble Alpes (UGA) Science and Technology

Research Internship Apr 2025 – Jul 2025

Universidad de Guadalajara (UdeG), CUCEI Education

Science Outreach: Jan 2017 – Dec 2019

- **Noche de las Estrellas (2017, 2018, 2019)**
- **Circo de la Ciencia**
- **Others**

Presented experiments and explained scientific phenomena to encourage **community engagement** with science; answered questions from observers.

Hewlett-Packard (HP) Education

Hour of Code Jan 2020
Partnered with local schools to teach coding basics to young girls and underrepresented groups, helping inspire the next generation of reinventors.

Agencia Espacial Mexicana (AEM) Science and Technology

Student Volunteer Sep 2016
Assistant at the 67th International Astronautical Congress (IAC) 2016.

Hobbies & Interests

- **Resilience & Well-being:** Mindfulness, swimming, cycling, running, and gym training.
- **Creativity & Innovation:** Photography and research in biology, physics, mathematics, and computer science.
- **Learning & Exploration:** Traveling, Chinese, philosophy, financial literacy and personal investments.
- **Community Engagement:** Science outreach, volunteering, and mentoring.

Last updated: October 11, 2026